

HOVDEN-LUNDANE, Norway

WMO: 014410

Lat: 59.5767N

Lon: 7.3897E

Elev: 2743

StdP: 13.30

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.4	-10.8	-20.3	2.0	-15.3	-15.4	2.6	-10.5	23.0	28.4	21.1	27.5	3.7	40	0.741

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	70.6	55.7	67.3	54.2	63.6	53.0	57.8	66.7	56.1	63.9	54.2	61.4	7.3	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.0	68.5	60.5	52.2	64.1	59.2	50.5	60.2	57.2	26.0	66.6	24.9	63.6	23.7	61.6	64.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	16.6	14.4	-20.9	74.6	4.5	4.0	-24.1	77.5	-26.8	79.9	-29.3	82.1	-32.6	85.0
			-21.2	59.8	4.5	2.3	-24.4	61.4	-27.1	62.7	-29.6	64.0	-32.8	65.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.1	18.6	18.4	23.6	31.6	39.5	46.4	51.2	49.7	44.4	35.7	27.4	21.0
	DBStd	14.38	11.70	11.70	9.84	7.06	6.16	4.79	5.07	4.54	5.26	6.73	9.36	12.44
	HDD50	5996	972	886	820	551	331	128	45	63	181	443	678	898
	HDD65	11295	1437	1306	1285	1001	790	558	426	474	619	908	1128	1363
	CDD50	175	0	0	0	0	6	20	84	54	11	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	14	0	0	0	0	0	0	14	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.2	7.1	6.6	6.7	5.8	5.8	6.0	5.6	5.5	6.1	6.0	6.4	6.6
Precipitation	PrecAvg	43.3	4.9	3.8	3.3	2.1	1.9	2.5	3.0	3.7	3.5	4.3	4.3	4.9
	PrecMax	62.8	13.3	9.6	13.2	3.5	4.9	5.8	5.6	7.2	6.7	8.4	8.8	12.4
	PrecMin	24.2	0.3	0.4	0.0	0.5	0.1	0.4	0.6	0.5	1.4	0.6	1.1	0.2
	PrecStd	8.4	3.0	2.6	2.5	0.7	1.0	1.4	1.0	1.5	1.4	2.2	2.2	2.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.7	40.9	45.7	53.2	68.8	71.6	76.0	70.7	65.0	54.5	46.4	41.7
		MCWB	36.3	37.5	37.6	43.3	52.2	52.3	58.3	56.5	53.0	45.6	43.6	39.3
	2%	DB	36.9	37.6	40.5	48.6	62.5	68.0	72.1	66.6	60.3	50.8	43.8	39.6
		MCWB	34.8	35.3	35.9	40.4	50.4	52.1	57.5	55.1	51.0	45.3	41.4	37.6
	5%	DB	35.2	35.7	38.1	44.8	56.8	63.7	68.9	63.2	56.8	48.2	41.1	37.5
		MCWB	33.3	33.5	34.8	38.4	46.8	50.5	56.1	54.0	50.4	44.2	38.8	35.7
	10%	DB	33.3	33.4	36.2	41.8	52.1	59.5	64.9	60.1	53.9	45.6	38.7	35.4
		MCWB	31.6	31.4	33.2	36.7	44.2	48.6	54.5	52.3	48.8	42.2	36.5	33.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.9	38.0	39.0	44.2	54.1	55.7	60.7	58.4	56.1	48.7	44.0	39.6
		MCDB	38.5	40.4	42.6	50.3	64.7	64.3	71.0	66.9	62.3	51.7	46.0	41.1
	2%	WB	35.2	35.7	36.6	41.4	50.7	54.0	58.8	56.8	52.9	46.4	41.4	37.7
		MCDB	36.8	37.6	39.5	46.9	60.5	64.3	69.5	63.2	57.2	49.7	43.7	39.3
	5%	WB	33.5	33.7	35.3	39.3	47.7	52.0	57.1	55.2	51.0	44.7	39.0	35.7
		MCDB	35.1	35.5	38.0	44.3	55.1	60.4	66.3	61.5	55.6	47.7	41.0	37.2
	10%	WB	31.7	31.5	33.2	37.3	45.1	50.0	55.3	53.3	49.3	42.4	36.7	33.7
		MCDB	33.2	33.2	35.8	41.1	51.7	57.6	63.6	59.2	53.6	45.2	38.6	35.3
Mean Daily Temperature Range	5% DB	MDBR	13.9	15.9	18.2	16.8	16.8	20.2	18.4	17.3	15.5	13.7	12.3	14.0
		MCDBR	12.2	13.0	16.8	20.8	26.4	31.5	30.0	24.1	22.6	18.2	11.6	10.1
		MCWBR	11.3	11.2	12.5	13.6	15.4	17.2	16.0	14.0	14.4	12.9	10.2	9.2
	5% WB	MCDBR	12.2	12.5	14.8	17.7	23.9	24.3	25.3	20.8	17.9	13.6	11.1	10.7
		MCWBR	11.7	11.1	11.2	11.7	14.5	14.6	14.0	12.8	12.4	10.8	9.9	9.9
Clear-Sky Solar Irradiance	taub		0.219	0.255	0.285	0.321	0.325	0.320	0.335	0.329	0.303	0.285	0.235	0.195
	taud		2.168	2.262	2.235	2.235	2.371	2.487	2.458	2.467	2.539	2.489	2.247	2.066
	Ebn at Noon		193	246	271	279	287	290	282	276	267	236	184	154
	Edh at Noon		13	21	30	36	34	31	31	29	23	17	12	10
All-Sky Solar Radiation	RadAvg		104	301	711	1289	1483	1589	1433	1124	720	352	132	68
	RadStd		24	72	152	247	263	198	194	98	75	60	27	10

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (55 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page